# Electric Field Redistribution via High-$k$ Dielectric Trench in GaN-on-Si AlGaN/GaN HEMTs

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We report a GaN-on-Si AlGaN/GaN high-electron-mobility transistor (HEMT) incorporating a drain-side $Pb(Zr,Ti)O_3$ (PZT) trench for electrostatic field management. The mechanism is formulated using the displacement-field boundary conditions and evaluated by two-dimensional TCAD with a field-independent relative-permittivity sweep from 3.9 to 500. The calculations reveal a nonmonotonic dependence of terminal voltage on permittivity and a broader drain-side high-field distribution for the nominal $\varepsilon_r = 300$ case. Experimentally, a representative PZT-trench HEMT reaches 1405 V at $L_{\mathrm{GD}} = 9$ $\mu$m, compared with 783 V for a co-fabricated conventional device and 957 V for a previously reported hybrid Schottky–ohmic drain HEMT, corresponding to breakdown-voltage improvements of 79% and 47%, respectively, while retaining a specific ON-resistance of 1.47 m$\Omega\cdot$cm$^2$. The dynamic-to-static ON-resistance ratios, $R_{\mathrm{ON,dyn}}/R_{\mathrm{ON,static}}$, are 1.6× and 2.5× after 900 and 1200 V off-state stress, respectively, at a 3 $\mu$s extraction delay. These results demonstrate simultaneous enhancement of voltage handling and dynamic performance through drain-side dielectric-boundary engineering.

High-electron-mobility transistors fabricated on GaN-on-Si substrates are attractive for high-voltage power applications because of their low cost, large wafer size, and compatibility with mature silicon processing technologies[1]. However, the breakdown voltage of conventional GaN-on-Si AlGaN/GaN HEMTs remains lower than that of devices on SiC substrates, particularly for structures with thin epitaxial layers ($\leq 4.0$ $\mu$m)[2]. A key limitation is severe electric-field crowding in the gate–drain region, especially at the drain-side edge under off-state bias. Although surface passivation, field plates, buffer optimization, and advanced GaN architectures have been widely adopted to improve breakdown performance[3–9], the drain-side peak field can still promote hot-electron injection and trap-related carrier capture. High-$k$ dielectric RESURF and polarization-superjunction concepts offer a complementary route for electrostatic field management and have enabled device designs approaching material limits and multi-kilovolt voltage scaling[10–13]. Here, a PZT-filled drain-side trench is experimentally integrated into a lateral GaN-on-Si HEMT. The central advance is the combined improvement of breakdown voltage and dynamic ON-resistance in a co-fabricated device pair, supported by a systematic TCAD permittivity sweep and two-dimensional field analysis.

The devices were fabricated on a 6-inch MOCVD-grown GaN-on-Si wafer comprising a 1.0 $\mu$m transition layer, a 3.0 $\mu$m C-doped GaN buffer, a 150 nm unintentionally doped GaN channel, a 3.0 nm AlN spacer, and a 20 nm $Al_{0.2}Ga_{0.8}N$ barrier. The conventional and PZT-trench HEMTs share the same epitaxy, $L_{\mathrm{G}} = 1.5$ $\mu$m, $L_{\mathrm{SG}} = 1.5$ $\mu$m, $L_{\mathrm{GD}} = 9$ $\mu$m, $L_{\mathrm{FP}} = 1$ $\mu$m, $W_{\mathrm{G}} = 100$ $\mu$m, passivation, and metallization. Following $F^+$ isolation and Ti/Al/Ni/Au ohmic formation, a drain-side trench with a lateral width of 1.5 $\mu$m and an etch depth of 3.0 $\mu$m was formed by $Cl_2/BCl_3/Ar$ ICP etching. $Pb(Zr_{0.53}Ti_{0.47})O_3$ was subsequently deposited by pulsed-laser deposition at approximately 600 °C to a nominal thickness of 3.0 $\mu$m, fully filling the trench, and annealed in $O_2$ at 600–650 °C for approximately 1 h. The principal dimensions and process conditions are summarized in Table I.

PZT is ferroelectric and its permittivity depends on frequency, field, thickness, and temperature[15–17]. The capacitance-derived trace used to select the simulation range decreases from approximately 400 at low frequency to approximately 100 near 1 MHz. Accordingly, $\varepsilon_r = 300$ is treated as an effective static TCAD parameter rather than a complete ferroelectric constitutive description. Beyond its high-$k$ response, PZT may modify local fields through polarization-bound interfacial charge. A polarization-induced field opposing drain-side field crowding could reduce the local peak; its effect depends on polarization orientation, domains, and screening. This contribution was not independently characterized under high-voltage operation, and the TCAD model excludes ferroelectric switching. A separate ferroelectric benefit therefore remains unverified. For the field-independent linear dielectric used in TCAD,

$$\nabla\cdot\mathbf{D} = \rho_{\mathrm{free}}, \qquad \mathbf{D} = \varepsilon_0\varepsilon_r\mathbf{E}, \qquad \mathbf{E} = -\nabla\phi, \\ \nabla\cdot\left[\varepsilon_0\varepsilon_r(\mathbf{r})\nabla\phi\right] = -\rho_{\mathrm{free}}(\mathbf{r}). \tag{1}$$

At a GaN/dielectric interface, the boundary conditions are

$$\hat{\mathbf{n}}\cdot\left(\mathbf{D}_{\mathrm{d}} - \mathbf{D}_{\mathrm{GaN}}\right) = \sigma_{\mathrm{f},int}, \qquad \hat{\mathbf{n}}\times\left(\mathbf{E}_{\mathrm{d}} - \mathbf{E}_{\mathrm{GaN}}\right) = 0. \tag{2}$$

Here, $\hat{\mathbf{n}}$ points from GaN to the dielectric and $\sigma_{\mathrm{f},int}$ denotes free interfacial sheet charge; for $\sigma_{\mathrm{f},int} = 0$,

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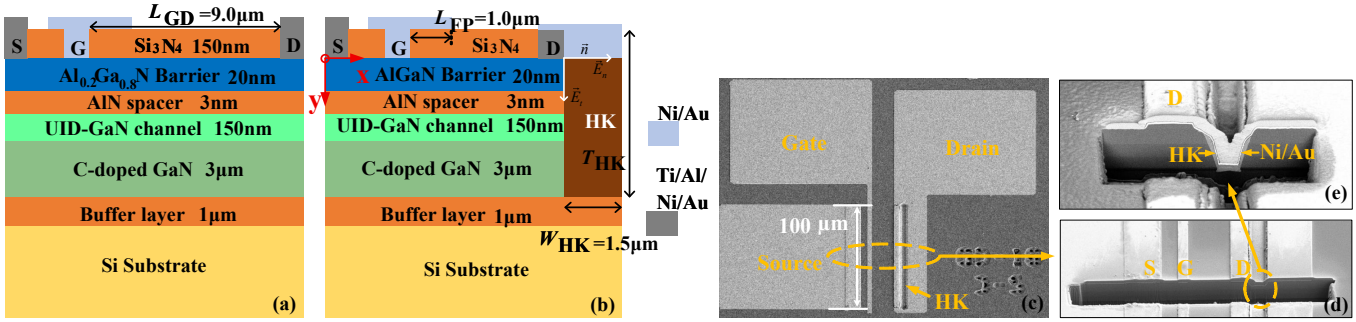



FIG. 1. Cross-sectional schematics of (a) the conventional and (b) the drain-side PZT-trench HEMTs. The arrows in (b) define the local interface-normal direction $\hat{\mathbf{n}}$ and the normal and tangential field components $E_n$ and $E_t$ used in Eq. (2). Electron microscopy of the fabricated structure: (c) top view, (d) source–gate–drain cross section, and (e) enlarged drain-side trench region.

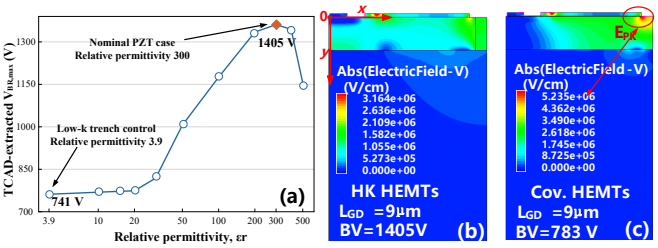



FIG. 2. (a) TCAD terminal-voltage metric versus trench relative permittivity. Simulated electric-field distributions near the respective terminal voltages of (b) the PZT-trench model at 1405 V and (c) the conventional model at 783 V, both at $V_{GS} = -7$ V. The independently autoscaled color bars report absolute field magnitude in V/cm and therefore permit quantitative comparison.

$\varepsilon_{r,d}E_{n,d} = \varepsilon_{r,GaN}E_{n,GaN}$ and $E_{t,d} = E_{t,GaN}$. Thus, the PZT trench redistributes the two-dimensional field through local displacement-field continuity.

Two-dimensional simulations were performed using Sentaurus SDE/SDevice V-2022.03-SP2. The numerical structure uses $L_{SG} = 1.5$ $\mu$m, $L_G = 1.5$ $\mu$m, $L_{GD} = 9$ $\mu$m, and a 1.5 $\mu$m × 3.0 $\mu$m dielectric trench. The trench permittivity is field independent and swept over $\varepsilon_r = 3.9$, 10, 15, 20, 30, 50, 100, 200, 300, 400, and 500, without ferroelectric switching or dielectric leakage. The model includes Fermi statistics, doping-dependent and high-field mobility, SRH recombination, electron avalanche generation, strain-dependent polarization, thermionic transport, and nonlocal electron tunneling. The GaN donor concentration is $2 \times 10^{16}$ cm$^{-3}$; the peak bulk deep-level concentration is $2 \times 10^{17}$ cm$^{-3}$; and the nitride/AlGaN and GaN/Si interface-trap densities are $5 \times 10^{13}$ cm$^{-2}$. All temperature-dependent calculations are isothermal.

The permittivity-dependent terminal-voltage metric in Fig. 2(a) is defined consistently for every simulated structure as

$$V_{BR,max}(\varepsilon_r) \equiv \max\{V_{DS} : \mathcal{S}(V_{DS}, \varepsilon_r) \text{ converges}, \; |I_D| < I_{stop}\}. \quad (3)$$

where $\mathcal{S}$ denotes the converged two-dimensional solution.

TABLE I. Device dimensions and fabrication parameters.

| Parameter | Nominal value or process |
|---|---|
| Transition buffer / C:GaN / UID-GaN | 1.0 / 3.0 / 0.15 $\mu$m |
| AlN / AlGaN | 3.0 / 20 nm |
| Al fraction in barrier | 0.20 |
| $Si_3N_4$ passivation | 150 nm, LPCVD |
| $L_G$ / $L_{SG}$ / $L_{GD}$ | 1.5 / 1.5 / 9.0 $\mu$m |
| $L_{FP}$ / $W_G$ | 1.0 / 100 $\mu$m |
| PZT trench $W_{HK} \times T_{HK}$ | $1.5 \times 3.0$ $\mu$m$^2$ |
| PZT deposited thickness | 3.0 $\mu$m |
| PZT target composition | $Pb(Zr_{0.53}Ti_{0.47})O_3$ |
| PLD / post-anneal | ∼ 600 °C / $O_2$, 600–650 °C, ∼ 1 h |
| Ohmic metallization | Ti/Al/Ni/Au, 900 °C, 30 s RTA |
| Gate metal | Ni/Au |

At fixed $V_{GS} = -7$ V, $V_{DS}$ is swept upward until $|I_D|$ reaches $I_{stop} = 10^{-10}$ A/$\mu$m. Equation (3) gives the highest converged sampled voltage below this threshold. The BV labels in the simulated field maps refer to these terminal voltages. The low-$k$ trench control ($\varepsilon_r = 3.9$) shares the geometry and device parameters of the nominal PZT case ($\varepsilon_r = 300$). Their terminal-voltage metrics are approximately 741 and 1405 V, respectively [Fig. 2(a)]. The same gate bias and extraction rule are used for both structures, providing a comparison of permittivity dependence within the linear-dielectric model. For the common trench geometry, $V_{BR,max}$ increases strongly as $\varepsilon_r$ rises from 3.9 to approximately 300 and then decreases at $\varepsilon_r = 500$. Thus, the simulated dependence is nonmonotonic, with the largest terminal-voltage metric occurring near $\varepsilon_r = 300$ within the examined parameter range; this nominal case is used for the field-distribution comparison below. To separate field crowding from the applied drain voltage, a dimensionless field-concentration factor is defined as

$$\Gamma(\varepsilon_r, V_{DS}) \equiv \frac{E_{pk,surf}(\varepsilon_r, V_{DS})}{V_{DS}/L_{GD}}. \quad (4)$$

Here, $E_{pk,surf}$ is extracted along the same GaN surface path. For comparisons at a specified drain bias, a smaller $\Gamma$ indicates reduced field crowding; $\Gamma$ is used only as an electrostatic diagnostic rather than as an independent

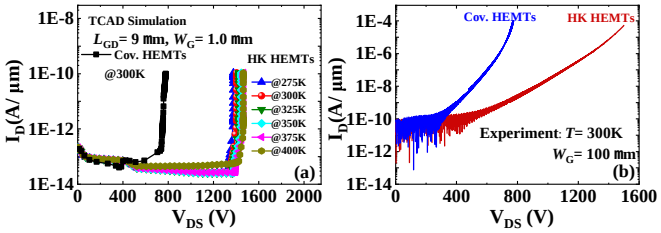



FIG. 3. Off-state characteristics: (a) isothermal TCAD results for the PZT-trench model from 275 to 400 K and the conventional model at 300 K; (b) room-temperature measurements of the co-fabricated conventional and PZT-trench HEMTs.

breakdown criterion. Figures 2(b) and 2(c) show that the PZT-trench model broadens the drain-side high-field region, while the conventional device retains a localized corner peak. At $V_{GS} = -7$ V, the conventional and PZT-trench models show drain-side surface peaks of approximately 5.1 and 1.7 MV/cm at their respective terminal voltages of 783 and 1405 V. The lower peak despite the higher PZT-trench voltage supports field redistribution at the respective terminal states, without establishing the experimental breakdown origin. For the nominal $\varepsilon_r = 300$ case, the field at the trench bottom is approximately 1.3 MV/cm and remains below the extracted drain-side surface peak.

Experimentally, at $L_{GD} = 9$ $\mu$m, the co-fabricated conventional and PZT-trench BVs are 783 and 1405 V [Fig. 3(b)], a 79% gain. For literature comparison, the measured BV of 1405 V is approximately 47% higher than the 957 V reported for the HSO HEMT in Ref.[14]. TCAD supports a field-redistribution interpretation of this enhancement; the exact experimental breakdown location was not identified. In the experiment, BV was extracted at $I_D = 10^{-4}$ A/$\mu$m for devices with $W_G = 100$ $\mu$m, corresponding to a total drain current of 10 mA. The simulated off-state sweeps use $V_G = -7$ V, whereas the room-temperature measurements use $V_{GS} = -5$ V with the source grounded. Because the two extraction criteria are not equivalent and the gate biases also differ, the TCAD results are used to compare terminal-voltage trends and electric-field redistribution rather than to quantitatively predict the absolute experimental BV. Figure 3(a) predicts a positive temperature coefficient for the PZT-trench model under isothermal simulation.

The more uniform equipotential spacing in Fig. 4(a), together with the lateral and vertical profiles in Figs. 4(d) and 4(e), is consistent with redistribution of the surface and buffer fields in the simulated structure. Because the profiles correspond to different drain-voltage sets, they describe near-terminal field evolution; Eq. (4) defines the normalized quantity to be used when comparing field crowding at a specified drain bias.

Figure 5(a) shows comparable output characteristics for the conventional, HSO, and PZT-trench devices, with threshold voltages of approximately $-2.5$ V. At $V_{GS} = 2$ V, their $R_{ON,sp}$ values are 1.43, 1.65, and 1.47 m$\Omega$·cm$^2$, respectively; the co-fabricated conventional and PZT-trench devices use the same $1.3 \times 10^{-5}$ cm$^2$

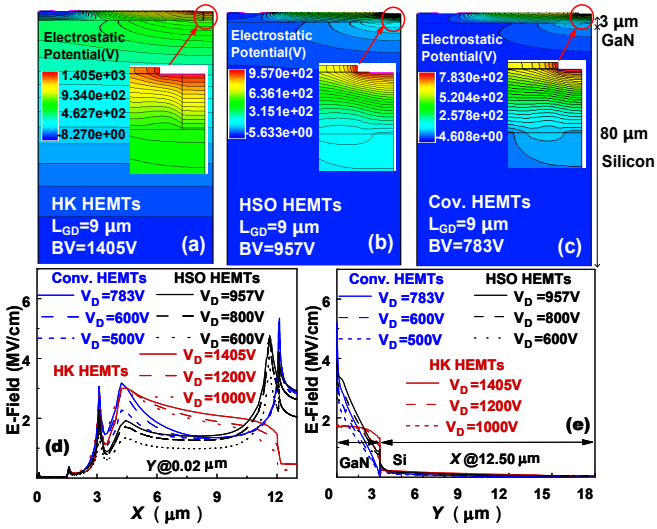



FIG. 4. Simulated electrostatic-potential contours near the respective terminal voltages of the (a) PZT-trench, (b) HSO, and (c) conventional structures; (d) lateral field at $Y = 0.02$ $\mu$m and (e) vertical field at $X = 12.50$ $\mu$m.

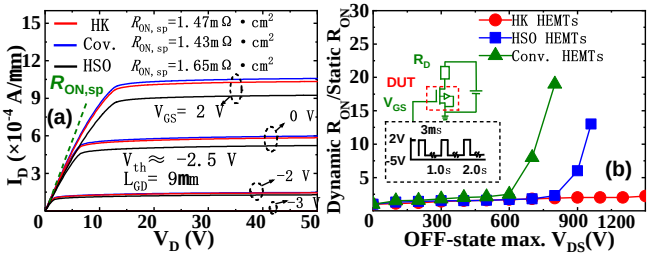



FIG. 5. (a) Measured output characteristics and specific ON-resistances. (b) Dynamic-to-static ON-resistance ratio versus off-state stress voltage, extracted 3 $\mu$s after the gate transition. Inset: switching circuit and gate-bias sequence.

active-area normalization including the trench footprint. The corresponding performance metrics are

$$R_{ON,sp} = A_{act} \left. \frac{V_D}{I_D} \right|_{lin}, \qquad BFOM = \frac{BV^2}{R_{ON,sp}}. \quad (5)$$

where the linear-region slope is used for $R_{ON,sp}$. The PZT-trench device yields a nominal BFOM of 1.34 GW/cm$^2$. Dynamic $R_{ON}$ was measured with a quiescent gate voltage of $-5$ V, an ON-state gate voltage of 2 V, and a 3 $\mu$s extraction delay. As shown in Fig. 5(b), the conventional and HSO curves increase sharply above approximately 700 and 900 V, respectively, whereas the PZT-trench device exhibits $R_{ON,dyn}/R_{ON,stat}$ ratios of 1.6 and 2.5 after 900 and 1200 V off-state stress. The reduced dispersion is consistent with mitigation of field-assisted trapping, although the measurement does not assign the response to a specific trap population.

Device-to-device statistics are unavailable, limiting the results to representative co-fabricated devices. Measurements across more devices and fabrication batches are needed to establish variability and reproducibility.

In conclusion, a drain-side PZT trench enables a representative GaN-on-Si HEMT to reach 1405 V at $L_{GD} = 9\ \mu$m while maintaining $R_{ON,sp} = 1.47\ \text{m}\Omega\cdot\text{cm}^2$ and moderated dynamic resistance degradation. The permittivity sweep and field distributions support dielectric-boundary-induced field redistribution as a plausible electrostatic interpretation of the observed improvement. Together, the experimental and numerical results demonstrate the potential of the PZT-trench architecture for field management in high-voltage lateral GaN devices.

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## AUTHOR DECLARATIONS

### Conflict of Interest

The authors have no conflicts to disclose.

### Author Contributions

**Zhen Cao:** Funding acquisition (lead); Project administration (lead); Resources (lead); Supervision (lead). **Zhiwei Hou:** Conceptualization (equal); Data curation (lead); Formal analysis (equal); Investigation (supporting); Writing – original draft (lead). **Qi Sun:** Formal analysis (supporting); Investigation (supporting); Methodology (lead). **Guohao Yu:** Formal analysis (supporting); Investigation (supporting). **Bingliang Zhang:** Investigation (lead); Formal analysis (supporting); Resources (supporting); Supervision (supporting). **Biao Hou:** Conceptualization (equal); Formal analysis (equal); Supervision (lead); Writing – review & editing (lead). **Licheng Jiao:** Supervision (supporting). **Yintang Yang:** Funding acquisition (lead); Supervision (lead); Writing – review & editing (supporting).

## DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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